

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

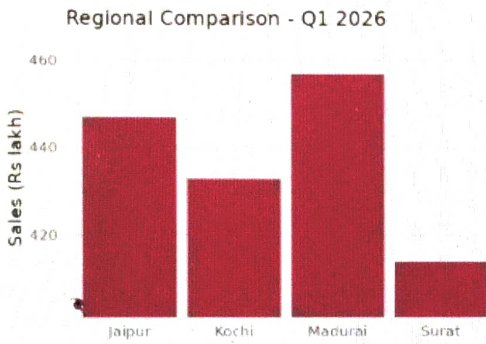


Chart 1

Budget Allocation by Department (14 categories)

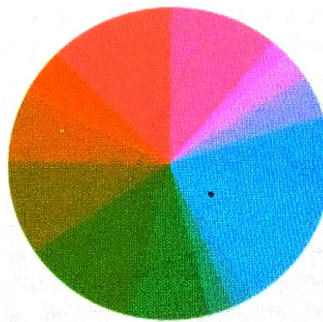


Chart 2

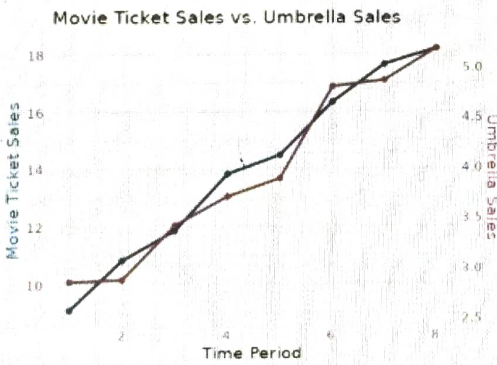


Chart 3

Study Hours vs. Exam Score

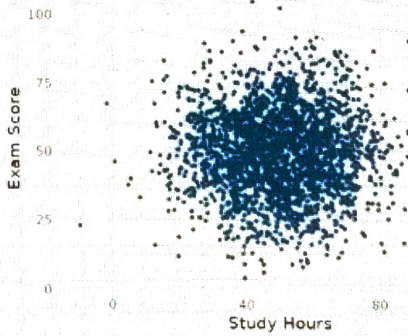


Chart 4

Date

01-10-2026

Session 1 - Activity

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Chart 1 - Truncated Y-axis CBar chart)

The chart compares sales across Jaipur, Kochi, Madurai, & Surat. Madurai appears to have much higher sales than the other regions, while Surat appears much lower.

However, the Y-axis begins around ₹400 lakh instead of ₹0. This truncates the baseline & visually the difference b/w the bars.

Rule: Bar charts should normally start at zero or use another because bar length represents the magnitude of the value.

Chart 2 - Too many categories in a pie chart

The chart represents budget allocation across 14 departments. It attempts to show each department contributes to the total budget.

However, the large no of slices makes the chart difficult to interpret. Several slices are similar in size, & comparing angles & areas is difficult.

Rule: Pie charts are most effective when there are only few categories.

Chart - 3 - Dual Y-axis, False correlation

The chart compares Movie Ticket Sales & Umbrella sales over diff time periods. If the two lines appear to move together, which may give the impression that the variables are related.

However, the two quantities use separate Y-axes with diff scales. The scales can be adjusted independently, making unrelated trends appear visually similar.

Rule: Dual-axis charts should be used carefully because independent scales can create misleading correlation.

Chart - 4 - Overplotting (Scatter plot)

The chart shows the relationship b/w study hours & Exam Score using a large no of observations.

The problem is that many points overlap because the points are fully opaque. Dense areas become dark solid regions.

Although the scatter plot gives an overall idea of the data.

Rule: Large datasets require techniques that make overlapping observations visible.